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Work Experience

Shikho Technologies

Dhaka, Bangladesh

Software Engineer (June, 2026 – Present)

- Designed and developed an **AI Agent pipeline** to dynamically generate personalized learning content (quizzes, model tests, and chapter summaries) from educational resources and lecture videos.
- Built and scaled the backend payment processing systems to handle transactions for over **2.5 million active users**, integrating local payment gateways and ensuring high availability.
- Collaborated with **Grameenphone (MyGP)** to integrate campaign features, enabling seamless user authentication and learning content delivery across platforms.
- Led database optimization and schema migrations for massive datasets, analyzing user data patterns and building real-time dashboards to improve system monitoring and business intelligence.

Cloudscocket

Boyd's, Maryland

Software Engineer (December, 2025 – August, 2026)

- Designed and built a scalable **billing and usage-metering platform** in Go supporting subscription- and usage-based pricing (plans, subscriptions, and usage metering) via an asynchronous data ingestion pipeline.
- Architected and deployed a decoupled, high-performance **asynchronous ML response streaming pipeline** using AWS SQS, AWS Lambda (with response streaming Function URLs), and Amazon ElastiCache Redis Streams, enabling sub-100ms streaming latency to frontend clients.
- Developed a robust, event-driven notification service using an asynchronous processing pipeline to dispatch real-time push notifications to mobile and web applications based on user activity triggers.

Technonext (US-Bangla Airlines Concern) *Software Engineer (April, 2025 – March, 2026)*

Dhaka, Bangladesh

- Led the backend team in building a distributed order-processing platform capable of handling over **10,000 order requests under peak load** through an asynchronous message-broker architecture and high-performance RPC communication. Designed for horizontal scalability and reliability, ensuring smooth order fulfillment during high-traffic events.
- Migrated **600,000+ geospatial records** from relational storage to a high-performance search engine, enabling **sub-100ms p99 latency** for geocoding and reverse-geocoding via optimized in-memory caching and indexed lookups.
- Developed a high-performance geo-service using open geospatial data and search indexing, delivering **sub-100ms average latency** for geocoding and reverse-geocoding—powering real-time location intelligence across multiple internal platforms.
- Participated in cross-functional teams for architectural and code reviews, contributed to decisions on database indexing, caching, and message-handling strategies to enhance system scalability and maintainability.
- Resolved a critical performance bottleneck in the monolithic inventory system by decoupling order processing into a dedicated microservice. Migrated REST-based inventory checks to RPC and integrated message broker for asynchronous messaging—reducing average response

time from **2.8s to 1.9s** (~32% improvement) under peak load.

Vivasoft Ltd.

Junior Software Engineer (July, 2023–March, 2025)

Dhaka, Bangladesh

- Built a scalable email gateway service and implemented a dynamic point-evaluation and ranking system, improving assessment accuracy from **76% to 95%** while reducing data-processing latency.
- Developed a middleware to convert markdown files into CMS pages, automated publication via CMS API and **reduced manual content processing by 90%**
- Deployed and managed backend applications on a cloud system using container orchestration, ensured **scalable, reliable**, and highly **available** services with **disaster recovery**, capable of handling high traffic and concurrent requests.
- Integrated a secure payment gateway, enabled reliable online transactions and streamlined the payment flow for end users that resulted in a **reduction in failed transactions and smoother checkout experience**.
- Re-architected the inventory management module into a real-time stock-tracking system across multiple warehouses, leveraged caching and database transactions reducing query times by **40%** and ensured consistency under concurrent updates.
- Designed and optimized database structures and implemented a secure file upload system with cloud storage support, reduced file-handling errors and improved **data accessibility** and retrieval speed for users.

Contributed Projects

Shikho

Backend Engineer

AI-driven content generation and personalized learning platform

Shikho Website

- Designed and developed a Go-based **AI Agent pipeline** to parse video lectures and textbooks, generating chapter summaries, model tests, and quizzes.
- Scaled high-throughput API endpoints and payment systems to support transactions and delivery for over **2.5 million active users**.

Foodi

Backend Engineer

Real-time inventory system for food delivery & supermarket platform

Foodi App

- Re-architected inventory module into a distributed, real-time tracking system, reducing stock discrepancies by **90%** during peak order surges using event-driven updates and atomic transactions.
- Decoupled inventory checks from monolithic flow into a dedicated microservice, migrating REST to **RPC** and integrating asynchronous messaging—cutting average sync latency from **2.8s to 1.9s** (~32% improvement).

Financfy

Backend Engineer

Cloud-based accounting software for SMEs

Financfy Website

- Engineered real-time inventory tracking system with multi-warehouse synchronization, reducing stock discrepancies by **90%** using event-driven updates.
- Integrated inventory module across distributed warehouses with **Redis** caching and **database transactions**, achieving **sub-50ms** stock availability checks.
- Collaborated with frontend to deliver accurate real-time inventory dashboards, improving operational decision-making speed.

Actionboard.ai

Backend Engineer

Enterprise AI workflow automation and subscription billing platform

Actionboard.ai Website

- Engineered a scalable **billing and usage-metering platform** in Go supporting subscription- and usage-based pricing, proration, and automated Stripe/SSLCommerz payments.
- Architected a decoupled **asynchronous ML response streaming pipeline** using AWS SQS, AWS Lambda (with response streaming Function URLs), and Redis Streams, delivering chunks to UI clients with sub-100ms latency.
- Built an event-driven notification service using RabbitMQ and Redis to dispatch real-time alerts across web and mobile platforms.

Mapnest

Backend Engineer

High-performance geocoding, reverse-geocoding & route snapping engine

- Migrated **600,000+ geospatial records** from relational storage to a high-performance search engine, enabling **sub-100ms p99 latency** for geocoding and reverse-geocoding via optimized in-memory caching and indexed lookups.
- Delivered **sub-100ms p99 latency** for **geocoding, reverse-geocoding, and snap-to-road** using high-speed spatial queries and in-memory caching—powering real-time location intelligence across internal platforms.

Wellteam

Backend Engineer

Online wellness platform with community programs

Wellteam Website

- Developed a scalable email gateway package, enabling reliable high-volume email delivery with retry mechanisms and delivery tracking.
- Led implementation of core point-evaluation and ranking logic, improving accuracy and reducing processing time by **40%** through optimized algorithms and concurrent processing.
- Integrated **Consul** for service discovery and **Stripe** for payments, ensuring resilient microservice communication and secure transaction handling.
- Containerized services with **Docker** and deployed on **AWS** infrastructure, enabling seamless scaling and zero-downtime updates.

Education

BSc.in Computer Science, Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh (2018–2023)

Technical Skills

Languages: Golang, TypeScript, Node.js, Python, JavaScript, C, C++ **Databases:** PostgreSQL, MySQL, MongoDB

Tools & Platforms: Git, Docker, Kubernetes, RabbitMQ, Redis, Elasticsearch

Cloud Platforms: AWS (EC2, S3, ECR, SES, SNS, SQS, Amplify, DynamoDB, Appsync)

Concepts: Clean Architecture, REST, gRPC, DDD, DevOps basics

Thesis / Projects

Vision Based Malware Classification Framework Based On Neural Network 2023

- **Book:** Vision Based Malware Classification Framework.pdf
- **Presentation:** Vision Based Malware Classification Framework.Presentation.pdf
- Converted malware binaries into images for neural network processing using the **Microsoft Malware Classification Challenge (BIG 2015)** database.
- Achieved **97.57% classification accuracy** using the CNN model.

Achievements

Anti-Corruption Commission National Debate Competition, Bangladesh 2015
National Champion

- Led a team that became **national champion** among **64 teams** from all districts of **Bangladesh**.

National Math Olympiad, Bangladesh 2010
Divisional Champion (Junior Category)

- Achieved **divisional champion** among **50–60 junior participants**.

Seasonal Competition, Bangladesh Shishu Academy 2014
Second Position, Division Level (Poem Recitation)

- Achieved **2nd place** among **8–10 participants**.